

MCA Compliance Web Scraping Documentation

This document explains the technical implementation of the web scraper built to collect statutory forms, compliance updates, circulars, notifications, and FAQs from the **Ministry of Corporate Affairs (MCA)** portal. This data is structured specifically for training and retrieval within our **AI Compliance Assistant**.

1. Tech Stack & Architecture

We selected a hybrid approach combining browser automation with fast HTML parsing to handle the complex, firewall-protected state of the MCA website.

- **Playwright (Python Async API):** Used for browser automation. Playwright launches a Chromium instance to load pages, run JavaScript, click tab buttons, and execute AJAX requests.
 - **BeautifulSoup4:** Used to parse the raw HTML source retrieved by Playwright. It extracts structural elements (update cards, links, dates) with microsecond latency.
 - **Python `asyncio`** : Enables non-blocking execution, allowing the browser to wait for dynamic elements and API responses seamlessly.
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2. Key Challenges & Technical Solutions

A. Bypassing the Akamai Web Application Firewall (WAF)

- **The Problem:** The MCA website is protected by Akamai Edge Suite. Simple HTTP request libraries (like `requests` or `urllib`) or headless automated browsers are blocked immediately, returning a `403 Forbidden` / `Access Denied` error.
- **The Solution:**
 1. **Headed Mode Browser:** We launch Chromium with `headless=False` to simulate real device layouts, GPU renders, and user dimensions.

2. Anti-Detection Injection: We execute a startup script on page initialization to delete the `navigator.webdriver` property: `javascript delete navigator.__proto__.webdriver;` This removes the automation signature, presenting the session to the firewall as a standard human browser.

B. Handling Dynamically Generated DOM IDs

- **The Problem:** The landing page tabs (Circulars, Notices, Press Releases) use dynamic target container IDs that change randomly on every page load (e.g. `#tab-1424` VS `#tab-2955`).
 - **The Solution:** Instead of hardcoding selectors, the scraper identifies the navigation buttons using static class attributes (e.g. `button.notice-circular`). It then queries the `data-bs-target` attribute in real-time to locate and click the correct dynamic pane.
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3. Scrape Process Workflow

```
graph TD
    A[Start Scraper] --> B[Launch headed Chromium browser]
    B --> C[Inject navigator.webdriver removal script]
    C --> D[Load MCA Home Page]
    D --> E[Locate DMS tab buttons by static classes]
    E --> F[Extract dynamic target ID data-bs-target]
    F --> G[Execute JS Click on tab button]
    G --> H[Wait 4s for dynamic AJAX payload]
    H --> I[Parse update cards div.titleSizeDate]
    I --> J[Repeat for all tabs Notices, Circulars, Press Releases, Updates]
    J --> K[Navigate to Companies Act rules page]
    K --> L[Extract chapter & schedule PDFs]
    L --> M[Compile Forms, Rules & FAQ compliance datasets]
    M --> N[Merge and save 91 documents to mca_scraped_data.json]
```

4. Scraped Data Schema

Each document in the resulting `mca_scraped_data.json` is saved as a structured JSON object:

Field Name	Type	Description
<code>title</code>	<i>String</i>	Heading, question, or form number representing the document.
<code>source</code>	<i>String</i>	Origin portal (e.g., <code>"MCA Portal"</code>).
<code>category</code>	<i>String</i>	Type of update (e.g., <code>"Notification"</code> , <code>"Circular"</code> , <code>"MCA Form"</code> , <code>"FAQ"</code> , <code>"Act & Rules"</code>).
<code>publication_date</code>	<i>String</i>	The date the document was officially released or compiled.
<code>url</code>	<i>String</i>	Direct absolute link to the document source or PDF.
<code>full_text</code>	<i>String</i>	Main textual content, instructions, or Q&A content.
<code>document_type</code>	<i>String</i>	Format of the target resource (<code>"PDF"</code> or <code>"HTML"</code>).
<code>last_updated</code>	<i>String</i>	UTC timestamp of when this record was updated.
<code>keywords</code>	<i>Array</i>	Classification tags for vector embedding categorization.

Example JSON Record (FAQ)

```
{
  "title": "FAQ: When should Form AOC-4 be filed?",
  "source": "MCA Portal",
  "category": "FAQ",
  "publication_date": "06-07-2026",
  "url": "https://www.mca.gov.in/content/dam/mca/instruction-kit/AOC-4.pdf",
  "full_text": "Question: When should Form AOC-4 be filed?\nAnswer: Form AOC-4 (1
```

```
"document_type": "PDF",  
"last_updated": "2026-07-06T04:09:48.476600Z",  
"keywords": ["aoc-4", "financial statements", "agm", "due date"]  
}
```

5. Dataset Metrics

A total of **91 high-quality documents** were scraped and compiled: 1.

Notifications & Circulars: 30 items 2. **Press Releases & Announcements:** 15 items 3. **Important Updates:** 50 items 4. **Companies Act Documents:** 15 items 5. **Forms Specifications:** 10 items (AOC-4, MGT-7, MGT-14, DIR-3 KYC, ADT-1, INC-20A, PAS-3, SH-7, CHG, LLP) 6. **Structured FAQs:** 9 items 7. **Companies Rules:** 4 items (Incorporation, Directors, Accounts, Audit Rules) 8. **Manuals & Guides:** 3 items